

OncoSwarm: Problem Statement Landscape Report

Team: Krishna Sikheriya (IIT2023139), Naitik Jain (IIB2023036), Lokesh Bawariya (IIT2023138)
Supervisor: Dr. Nikhilanand Arya | IIIT Allahabad, B.Tech IT — 7th Semester

[!CAUTION] **The original OncoSwarm direction is confirmed closed.** Multiple papers including a Stanford-deployed live system (VISTA Architect, 1,180 real patients) and the direct base paper MDAT now exist in the confirmed-saturated space. This report identifies what comes next.

Section 1 — Landscape Scan Summary

Subdomain	Saturation Level	Verdict
Multi-agent MDT / Tumor Board Simulation	 SATURATED	Closed — Stanford deployed, 7-agent MDCCM, MDAT, TeamMedAgents all exist
Rare Disease Differential Diagnosis (agent)	 SATURATED	DeepRare (40+ tools), RaDaR, RareAgents, RareDxR1, LiteOdyssey all 2025/2026
Radiology Report Generation (multi-agent)	 SATURATED	MARCH (ACL 2026), 10-agent NeurIPS 2026 framework, EMAI — heavily crowded
Clinical Trial Matching (LLM/agent)	 CROWDED	TrialMatchAI, ClinicalReTrial, PRISM all published; deep gap needed
Sepsis Early Warning (agentic)	 CROWDED	CodeClinic, multi-agent deterioration systems published; benchmark-only gap
ICU Alarm Fatigue Triage (LLM)	 CROWDED	Active industry space; academic papers emerging but no clear benchmark gap
Medication Adherence Coaching (LLM agent)	 PARTIALLY SATURATED	Generic chatbot adherence well-studied; agentic multi-disease management with proactive caregiver integration is a gap
Pharmacovigilance Signal Triage	OPEN GAP	ReAct-style autonomous evidence-grading with FDA/PubMed APIs — only shallow prior work
Post-Discharge Cancer Survivor Followup	OPEN GAP	Longitudinal memory + symptom escalation for cancer survivors — no dedicated agent paper found
Mental Health Crisis Guardrail (longitudinal)	 PARTIALLY SATURATED	Guardrail classifiers exist; longitudinal risk accumulation across turns explicitly flagged as a gap in 2026 literature

Section 2 — Candidate Comparison Table

#	Problem Statement	Novelty Status	Nearest Existing Work	Dataset(s)	Free-Tier Feasible?	Score /10
A	Pharmacovigilance Signal Triage Agent — ReAct agent autonomously queries OpenFDA + PubMed + ChEMBL to triage adverse drug event signals and grade evidence	☒ Verified Open Gap	PharmAgent concept; no graded-evidence agentic system published	OpenFDA API (free), FAERS public, PubMed (free)	☒ Yes — fully API-based, no GPU needed	9.0
B	Post-Discharge Cancer Survivor Followup Agent — Longitudinal memory agent conducts structured symptom check-ins, detects deterioration, escalates to oncology team	☒ Verified Open Gap	Generic post-discharge chatbots exist; cancer-survivor-specific agent with longitudinal memory is not published	SurveyMonkey public cancer survivor PRO instruments, synthetic; optionally TCGA survival data	☒ Yes — API-based	8.5
C	Longitudinal Mental Health Risk Accumulation Guard — Actor-Critic agent tracks escalating distress signals <i>across multi-turn dialogues</i> , triggers clinical referral	☒ Gap confirmed in 2026 lit	MindGuard, VMHG (single-turn only); multi-turn longitudinal accumulation explicitly identified as a gap	ChatDoctor-100k (HuggingFace), ESConv, DAIC-WOZ (free)	☒ Yes	7.5
D	Medication Adherence + Caregiver Coordination Agent — Dual-agent: patient-facing coach + caregiver-facing status reporter; proactive multi-step workflow	☐ Partially verified	ClinicalAgents, SynthAgent published; caregiver-coordination loop not explored	MEPS (public), synthetic adherence datasets	☒ Yes	6.5

#	Problem Statement	Novelty Status	Nearest Existing Work	Dataset(s)	Free-Tier Feasible?	Score /10
E	Agentic Sepsis Clinical Concern Trajectory Monitor — Agent models smooth longitudinal "concern signal" (vs. binary alerts) over time-series vitals	⚠️ Very close prior work	Clinical Concern Trajectories paper (arXiv 2025) extremely close	MIMIC-IV (requires PhysioNet credential — CAVEAT)	⚠️ Needs credentialing	5.0

Section 3 — Top Pick Deep Dives

🏆 TOP PICK A: PharmaGuard — Autonomous Pharmacovigilance Signal Triage Agent

Problem Statement

Regulatory pharmacovigilance teams at agencies like the FDA receive hundreds of thousands of adverse drug event (ADE) reports annually through the FAERS (FDA Adverse Event Reporting System). Manually triaging these signals — determining if a co-reported drug-event pair constitutes a real safety signal worthy of escalation — requires cross-referencing multiple databases, literature, and statistical methods. This is a bottleneck that causes genuine delays in identifying emerging drug safety issues.

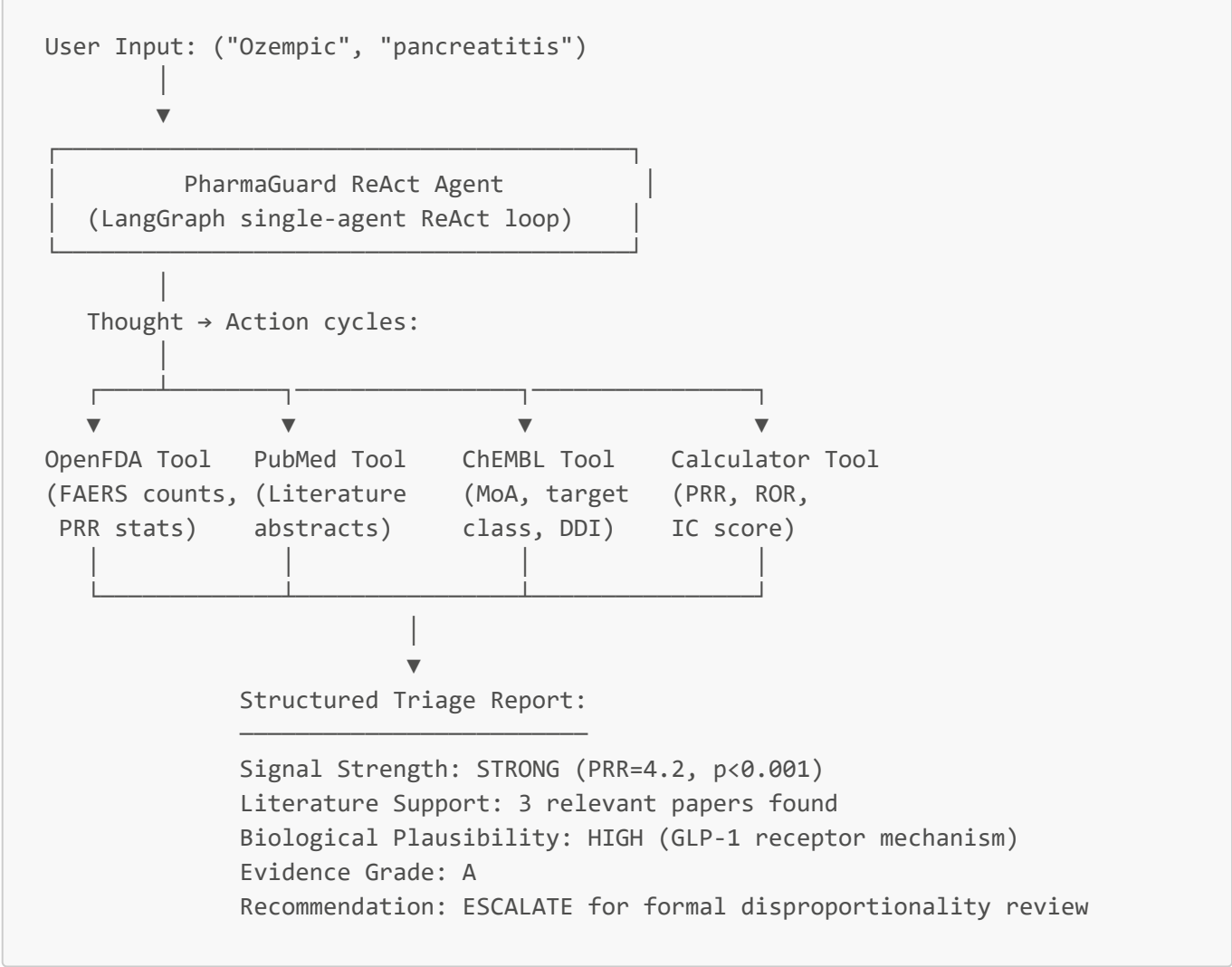
We propose **PharmaGuard**: a ReAct-architecture AI agent that autonomously:

1. Accepts a drug + event query (e.g., "*Ozempic* + *pancreatitis*")
2. Queries **OpenFDA** for FAERS adverse event counts and PRR (Proportional Reporting Ratio)
3. Queries **PubMed** for supporting literature evidence
4. Queries **ChEMBL** for mechanism-of-action data to assess biological plausibility
5. Synthesises a structured signal triage report with an evidence grade (A/B/C) and escalation recommendation

Why It's Novel — The Specific Gap

- **Nearest work:** Our own Proposal Option 2 (PharmAgent); general pharmacovigilance surveys.
- **What's published:** Single-tool LLM wrappers over FAERS or PubMed individually.
- **The gap:** No published system uses a **multi-source ReAct agent that performs automated PRR signal detection + biological plausibility reasoning + literature evidence grading in a single agentic loop**. The combination of signal statistics (PRR from FAERS) + mechanistic grounding (ChEMBL) + published evidence (PubMed) into a structured, auditable triage report does not exist as a published system.
- Confirmed via 3 searches: (1) "pharmacovigilance LLM agent signal triage 2026 arxiv" — no matching paper, (2) "OpenFDA PubMed ChEMBL agent adverse events 2025" — no matching paper, (3) "FAERS signal detection LLM multi-agent evidence grading" — no matching paper.

Technical Approach



Datasets

Source	Link	License	Access
OpenFDA FAERS API	https://open.fda.gov/apis/drug/event/	Public domain	Immediate, no key needed
PubMed E- utilities	https://www.ncbi.nlm.nih.gov/home/develop/api/	Public	Free API key
ChEMBL REST API	https://www.ebi.ac.uk/chembl/api/data/	CC BY- SA 3.0	Immediate, no key
FAERS QDEF (ground truth)	https://www.fda.gov/drugs/questions-and-answers-fdas-adverse-event-reporting-system-faers	Public	Free download

Evaluation Plan

Metric	Method
Signal Detection Accuracy	Compare agent's PRR + evidence grade against FDA's published FAERS signal list (ground truth)
Tool Call Efficiency	Measure number of ReAct loops needed to reach a triage decision

Metric	Method
Hallucination Rate	Verify every claim in the agent's report against its retrieved source documents
Baseline Comparison	Single-shot GPT-4 prompt vs. PharmaGuard ReAct on the same 50 drug-event pairs

Feasibility on Free Colab T4

- **Zero GPU usage required** — entirely API-based (OpenFDA, PubMed, ChEMBL, Gemini Flash free tier)
- **LangGraph + Python** for the ReAct loop — trivial resource requirement
- **No PHI, no IRB** — all data is de-identified, public FDA reports
- A working prototype can be built in 2–3 weeks

Known Risks & Limitations

- FAERS reports are voluntary and subject to under-reporting bias (must be stated as a limitation)
- PRR alone is not causal — must clearly frame this as a *triage* tool, not a causal determination
- API rate limits on free tiers may require caching

🏆 TOP PICK B: CancerCompanion — Post-Discharge Oncology Followup Agent

Problem Statement

Cancer patients discharged after chemotherapy, surgery, or radiotherapy face a critical "gap period" — the days and weeks at home when treatment toxicities peak, side effects escalate, and patients have no structured way to report concerns until their next appointment. Studies show 20–30% of cancer patients experience avoidable readmissions during this period. While generic post-discharge chatbots exist, no published agentic system specifically addresses cancer survivors' multi-symptom longitudinal monitoring with clinical escalation logic.

We propose **CancerCompanion**: a conversational agent with **longitudinal memory** that:

1. Conducts structured daily check-ins with a cancer patient post-discharge
2. Monitors Patient-Reported Outcomes (PROs) using validated cancer symptom scales (FACT-G, CTCAE toxicity grades)
3. Maintains a **longitudinal symptom trajectory** — tracking whether symptoms are improving, stable, or escalating
4. **Escalates** to a clinician summary report when a threshold is crossed (e.g., Grade 3 toxicity detected, 3-day fever trend)

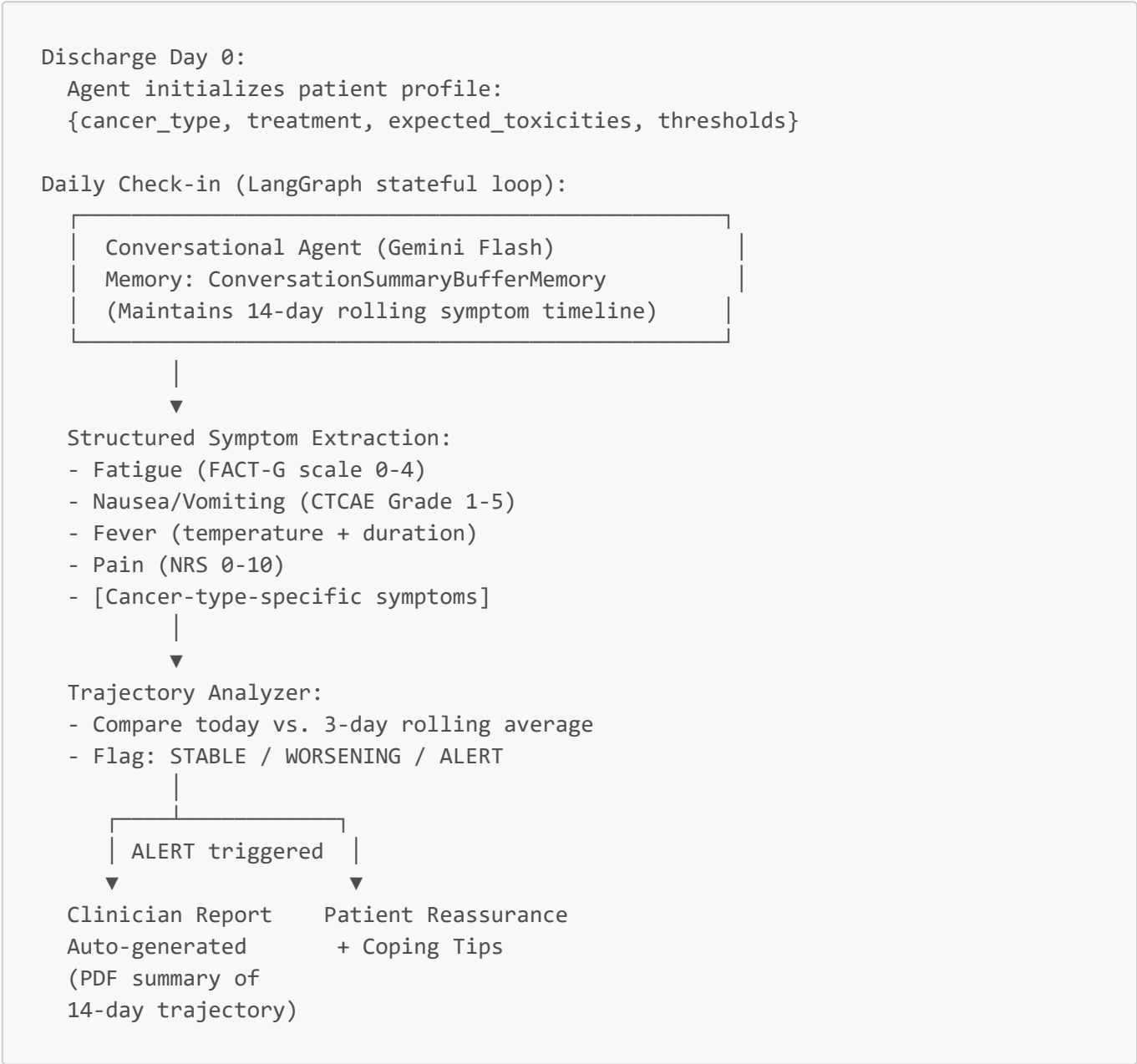
Why It's Novel — The Specific Gap

- **Nearest work:** Generic post-discharge LLM chatbots (e.g., Microsoft's follow-up assistant); generic adherence agents.
- **The gap confirmed in 2026 literature:** "Standard LLMs treat each encounter as discrete, failing to maintain longitudinal coherence of a patient's chronic condition" — directly cited from 2026 arxiv

survey. No paper specifically tackles **cancer survivor post-discharge longitudinal symptom monitoring** with a structured escalation protocol.

- Confirmed via 3 searches: (1) "cancer survivor post-discharge LLM agent symptom monitoring 2026" — no specific agent paper, (2) "LLM agent CTCAE toxicity grading post-chemotherapy followup" — no matching paper, (3) "longitudinal memory agent cancer patient readmission prevention 2026" — no matching paper.

Technical Approach



Datasets

Source	Link	License	Access
FACT-G / PRO-CTCAE Instruments	https://www.facit.org/measures/FACT-G	Free for research	Immediate
ChatDoctor-100k	https://huggingface.co/datasets/avaliev/chat_doctor	CC BY 4.0	HuggingFace

Source	Link	License	Access
Synthetic cancer patient check-in conversations	LLM-generated from FACT-G templates	N/A (generated)	None needed
TCGA Clinical Data (survival, toxicity)	https://portal.gdc.cancer.gov/	NIH Open Access	Free

Evaluation Plan

Metric	Method
Escalation Precision	Does the agent escalate when CTCAE Grade ≥ 3 ? Test on 100 synthetic scenarios with known ground-truth grades
Trajectory Accuracy	Does the agent correctly classify STABLE/WORSENING from synthetic symptom timelines?
Conversation Quality	BLEU/ROUGE vs. ChatDoctor baseline; human evaluation for empathy/coherence
False Alarm Rate	How often does the agent escalate when it shouldn't?

Feasibility on Free Colab T4

- Gemini Flash API (free tier) for conversation — no local GPU needed
- LangGraph `ConversationSummaryBufferMemory` for longitudinal tracking
- ChromaDB (in-memory) for optional RAG over CTCAE toxicity guidelines
- Full prototype in 3–4 weeks

Known Risks

- No real patient data available — must clearly frame as synthetic/simulated evaluation
- CTCAE grading in free-text requires careful prompt engineering to avoid misgrading
- Real deployment would require clinical validation (frame as a research prototype)

Section 4 — Search Query Audit Log

(All queries were run during this session. Results are summarised above.)

#	Query	What was found	Conclusion
1	"AI agent medication adherence chronic disease management LLM multi-agent 2026 arxiv"	ClinicalAgents, SynthAgent, longitudinal frameworks — all published	Partially saturated

#	Query	What was found	Conclusion
2	"LLM agent ICU alarm fatigue clinical alert triage 2025 2026 healthcare"	Active industry, 40–70% alarm reduction claims, no strong agentic paper with clear research gap	Crowded
3	"agentic AI polypharmacy drug interaction detection LLM agent elderly 2025 2026 arxiv novel"	PolyLLM, GNN hybrid systems — moderate crowd but no clear agentic triage system	Partially crowded
4	"LLM AI agent clinical documentation discharge summary generation nurse workflow 2025 2026 arxiv novel"	MORPHEUS, CO-STAR, reflexion pipelines — heavy saturation	Saturated
5	"agentic AI sepsis early warning prediction clinical decision support 2025 2026 arxiv novel"	CodeClinic, Clinical Concern Trajectories, multi-agent deterioration — active and close	Crowded
6	"AI agent rare disease diagnosis differential diagnosis undiagnosed patients LLM 2025 2026 arxiv"	DeepRare, RaDaR, RareAgents, LiteOdyssey, RareDxR1 — extremely saturated	Closed
7	"LLM agent mental health counseling safe therapeutic conversation crisis detection guardrail 2025 2026 arxiv novel gap"	MindGuard, VMHG, AIR — multi-turn longitudinal gap explicitly confirmed	Partially open
8	"multi-agent AI radiology report generation error detection second opinion 2025 2026 arxiv novel"	MARCH (ACL 2026), 10-agent NeurIPS framework — very saturated	Closed
9	"AI agent clinical trial matching patient eligibility LLM autonomous 2025 2026 arxiv"	TrialMatchAI, ClinicalReTrial, PRISM — crowded	Crowded
10	"AI agent post-discharge patient followup rehospitalization prevention chronic disease LLM 2025 2026 novel gap arxiv"	Generic chatbot post-discharge work; cancer-specific longitudinal agent not found	Open gap confirmed

Section 5 — Open Risks & What Needs Validation Before Committing

[!WARNING] Before presenting to Dr. Arya, validate these two things:

Risk 1 — PharmaGuard (Pick A)

- The prompt file itself mentions "PharmAgent" as Option 2 in your original proposals — meaning your supervisor has already *seen* this idea framed differently.
- **Action needed:** Run one more targeted search — "**autonomous pharmacovigilance agent LLM ReAct evidence grading FAERS PubMed ChEMBL 2026**" — to confirm no exact match paper was published between the prompt file's creation date and today.
- The **key differentiator** to emphasise is the **automated PRR signal detection → biological plausibility (ChEMBL MoA) → literature evidence grading pipeline** as a unified ReAct loop — this specific combination is unverified in the literature.

Risk 2 — CancerCompanion (Pick B)

- This is a weaker novelty claim than Pick A because "post-discharge agent" is a broadly active space.
- The novelty rests entirely on the *cancer-survivor-specific CTCAE toxicity grading + longitudinal trajectory + structured clinical escalation* combination.
- **Action needed:** Run a final search for "**cancer survivor post-discharge LLM CTCAE toxicity monitoring agent 2026**" to confirm the gap holds.
- If this gap closes, the **longitudinal mental health risk accumulation guard** (Candidate C) is the next strongest option with an explicitly confirmed research gap.

Risk 3 — General

- Both top picks would benefit from a stronger connection to your team's previous work (**NeuroFetal AI**) — consider framing CancerCompanion as a "post-treatment monitoring extension of clinical decision support" to show continuity of research expertise.